

Лабораторная работа №1

Тема: Освоение инструментария для выполнения работ, построение простой сети

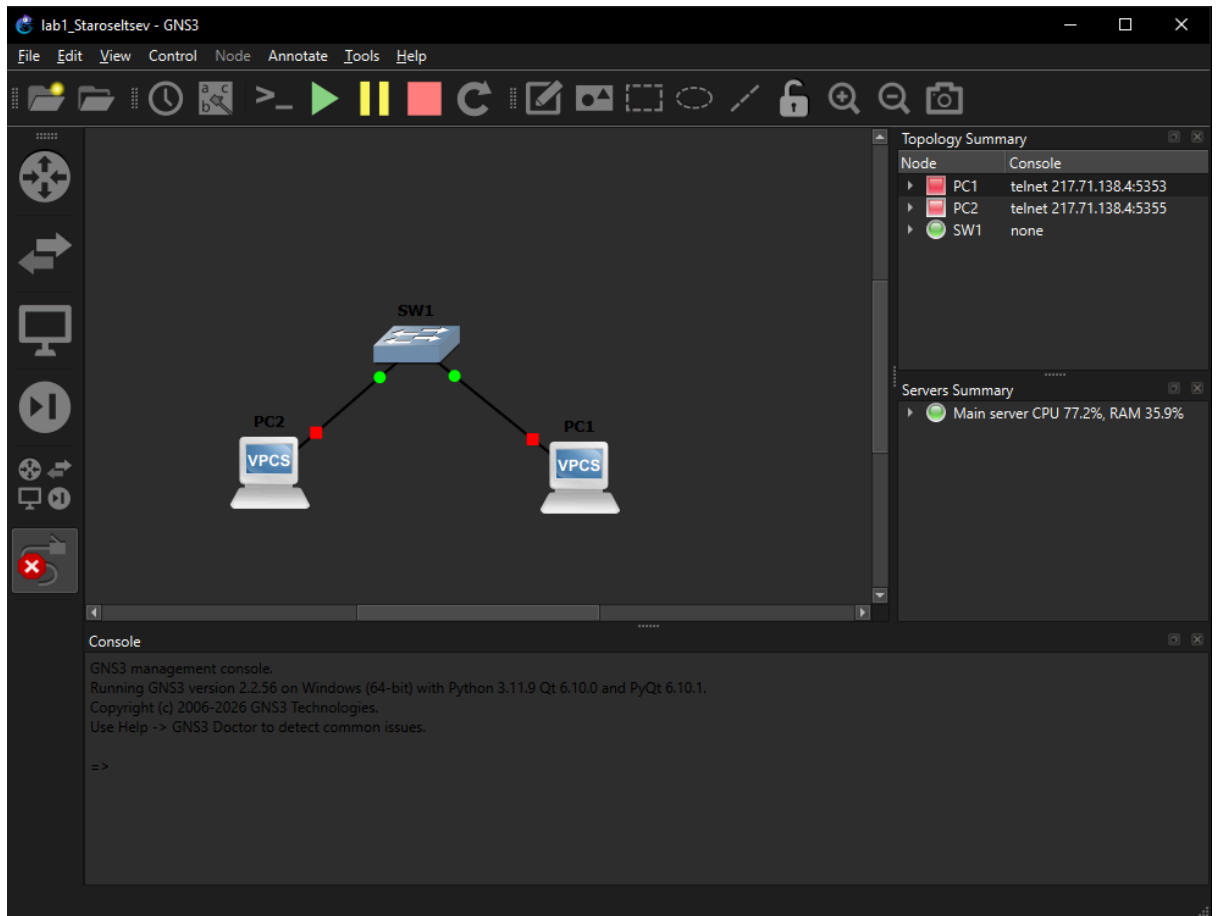
Все команды для настройки включаются в отчет в текстовом виде, не скриншоты.

nb! - отметка в тексте, "обратите особое внимание"

- 1) Установить и настроить эмулятор GNS3
- 2) Создать простейшую сеть, состоящую из 1 коммутатора и 2 компьютеров, назначить им произвольные ip адреса из одной сети
- 3) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера
- 4) Перехватить трафик протокола arp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark
- 5) Создать простейшую сеть, состоящую из 1 маршрутизатора и 2 компьютеров, назначить им произвольные ip адреса из разных сетей
- 6) Запустить симуляцию, выполнить команду ping с одного из компьютеров, используя ip адрес второго компьютера
- 7) Перехватить трафик протокола arp и icmp на всех линках(nb!), задокументировать и проанализировать заголовки пакетов в программе Wireshark, для фильтрации трафика, относящегося к указанному протоколу использовать фильтры Wireshark

1. Установил GNS3 и подключился к серверу

2. Собрал сеть



Взял сеть 192.168.10.0/24

Выбрал айпи для компьютеров:

PC1: 192.168.10.1/24

PC2: 192.168.10.2/24

ip 192.168.10.1/24

show ip

```
PC1> ip 192.168.10.1/24
Checking for duplicate address...
PC1 : 192.168.10.1 255.255.255.0

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.10.1/24
GATEWAY    : 0.0.0.0
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 21742
RHOST:PORT : 127.0.0.1:21743
MTU        : 1500
```

ip 192.168.10.2/24

show ip

```
PC2> ip 192.168.10.2/24
Checking for duplicate address...
PC2 : 192.168.10.2 255.255.255.0

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.10.2/24
GATEWAY     : 0.0.0.0
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 21744
RHOST:PORT  : 127.0.0.1:21745
MTU         : 1500
```

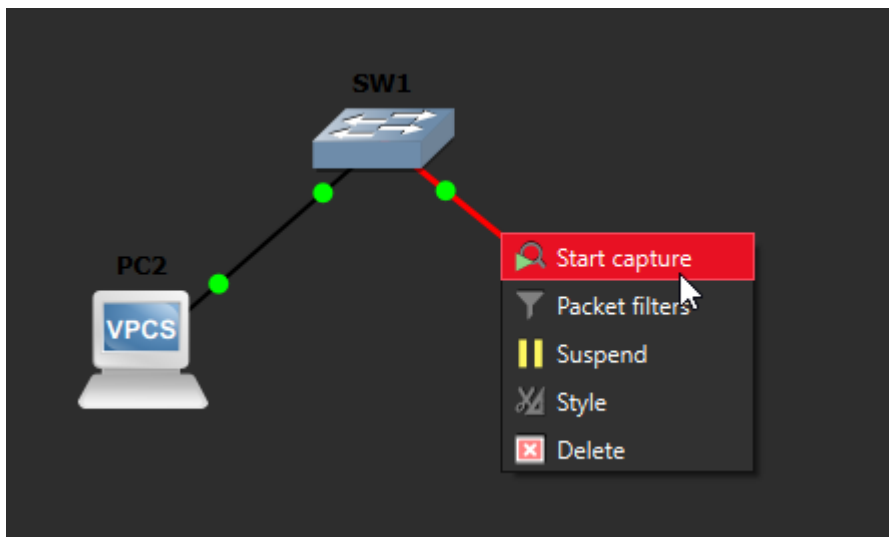
3. Пинг с PC1 до PC2

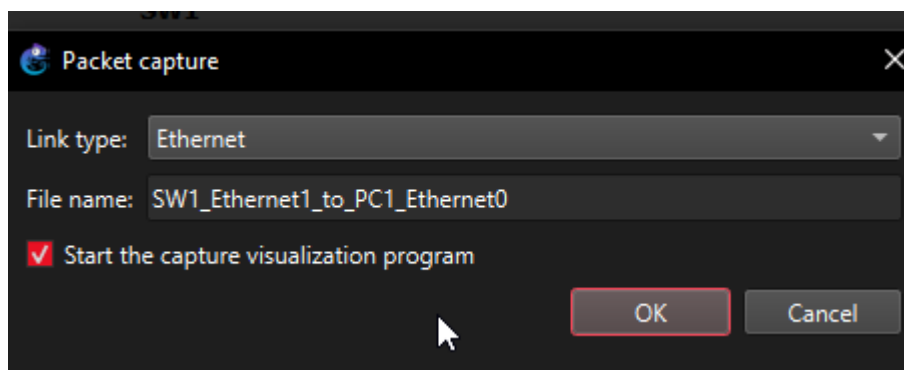
ping 192.168.10.2

```
PC1> ping 192.168.10.2

84 bytes from 192.168.10.2 icmp_seq=1 ttl=64 time=0.371 ms
84 bytes from 192.168.10.2 icmp_seq=2 ttl=64 time=0.351 ms
84 bytes from 192.168.10.2 icmp_seq=3 ttl=64 time=0.178 ms
84 bytes from 192.168.10.2 icmp_seq=4 ttl=64 time=0.184 ms
84 bytes from 192.168.10.2 icmp_seq=5 ttl=64 time=0.258 ms
```

4.

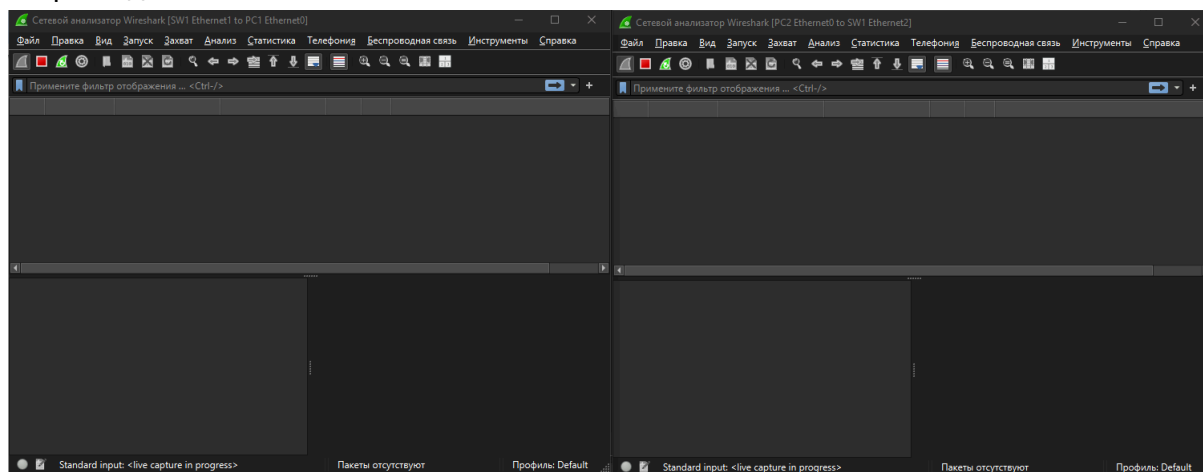




Открылся Wireshark

Аналогично для второго линка

Открыто два окна Wireshark

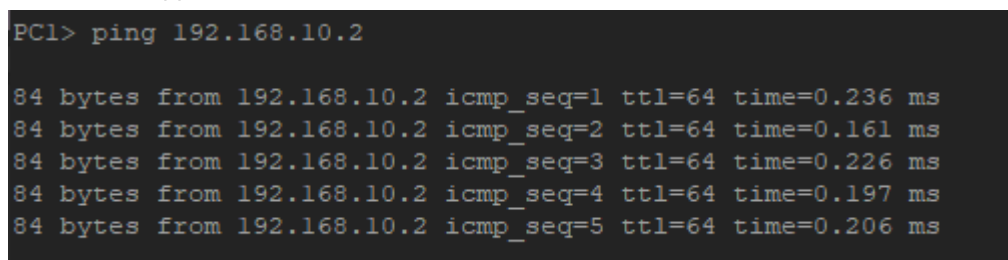


На обоих компьютерах очистил кэш arp



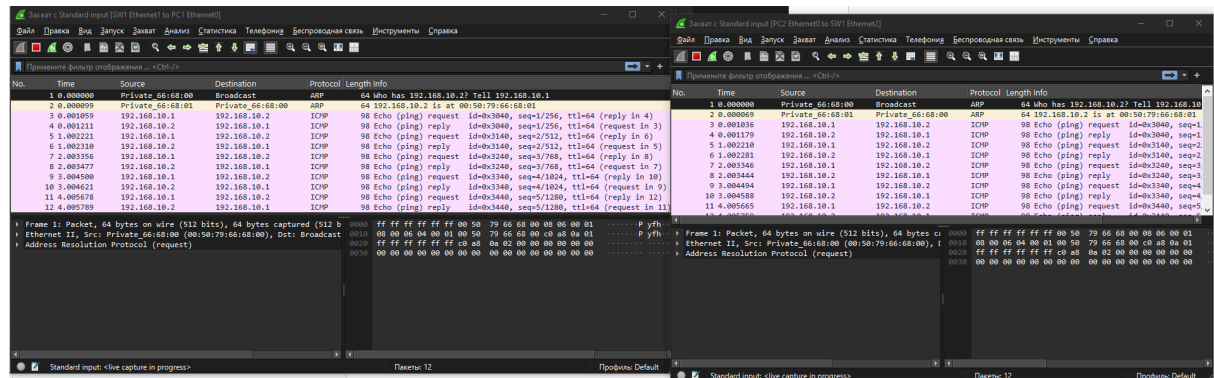
clear arp

Пинг с PC1 до PC2

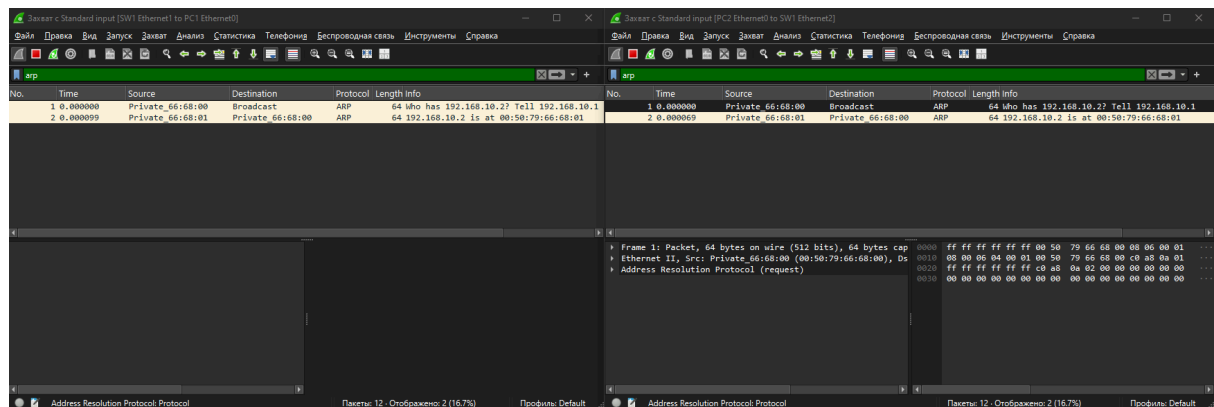


ping 192.168.10.2

Захваченные пакеты Wireshark'ом

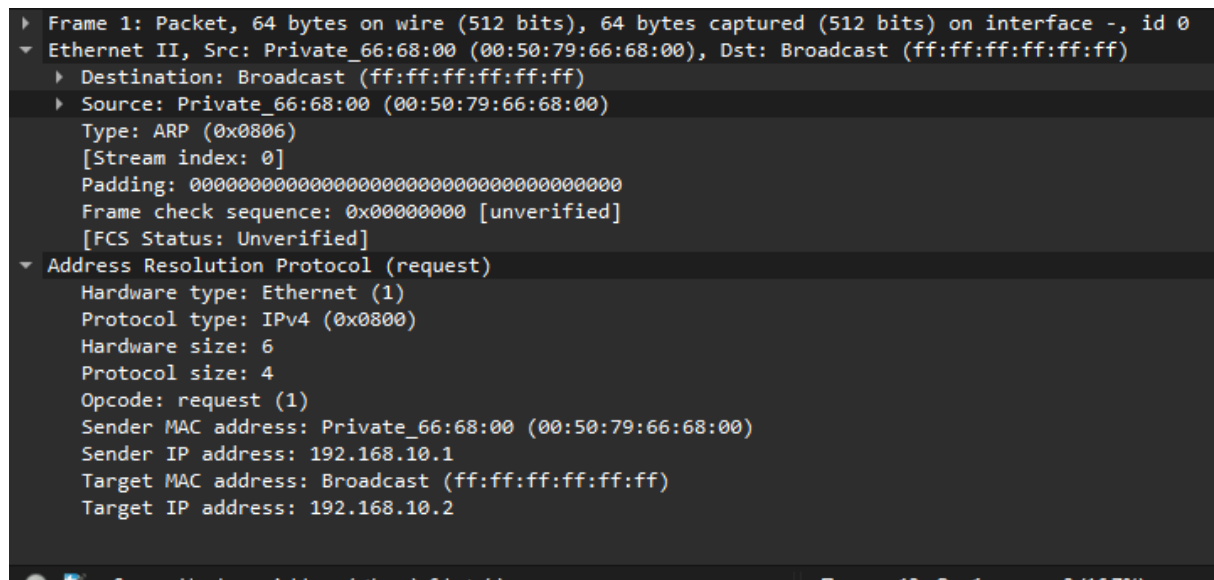


Отфильтрованные arp



Анализ пакетов:

Запрос



Ethernet II:

Destination MAC: ff:ff:ff:ff:ff:ff

Source MAC: 00:50:79:66:68:00

Type: ARP

Address Resolution Protocol:

Hardware type: Ethernet
Protocol type: IPv4
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: 00:50:79:66:68:00
Sender IP address: 192.168.10.1
Target MAC address: 00:00:00:00:00:00
Target IP address: 192.168.10.2

Ответ:

```
▶ Frame 2: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
▼ Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Private_66:68:00 (00:50:79:66:68:00)
  ▶ Destination: Private_66:68:00 (00:50:79:66:68:00)
  ▼ Source: Private_66:68:01 (00:50:79:66:68:01)
    .... ..0. .... = LG bit: Globally unique address (factory default)
    .... ..0. .... = IG bit: Individual address (unicast)
  Type: ARP (0x0806)
  [Stream index: 1]
  Padding: 00000000000000000000000000000000
  Frame check sequence: 0x00000000 [unverified]
  [FCS Status: Unverified]
▼ Address Resolution Protocol (reply)
  Hardware type: Ethernet (1)
  Protocol type: IPv4 (0x0800)
  Hardware size: 6
  Protocol size: 4
  Opcode: reply (2)
  Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
  Sender IP address: 192.168.10.2
  Target MAC address: Private_66:68:00 (00:50:79:66:68:00)
  Target IP address: 192.168.10.1
```

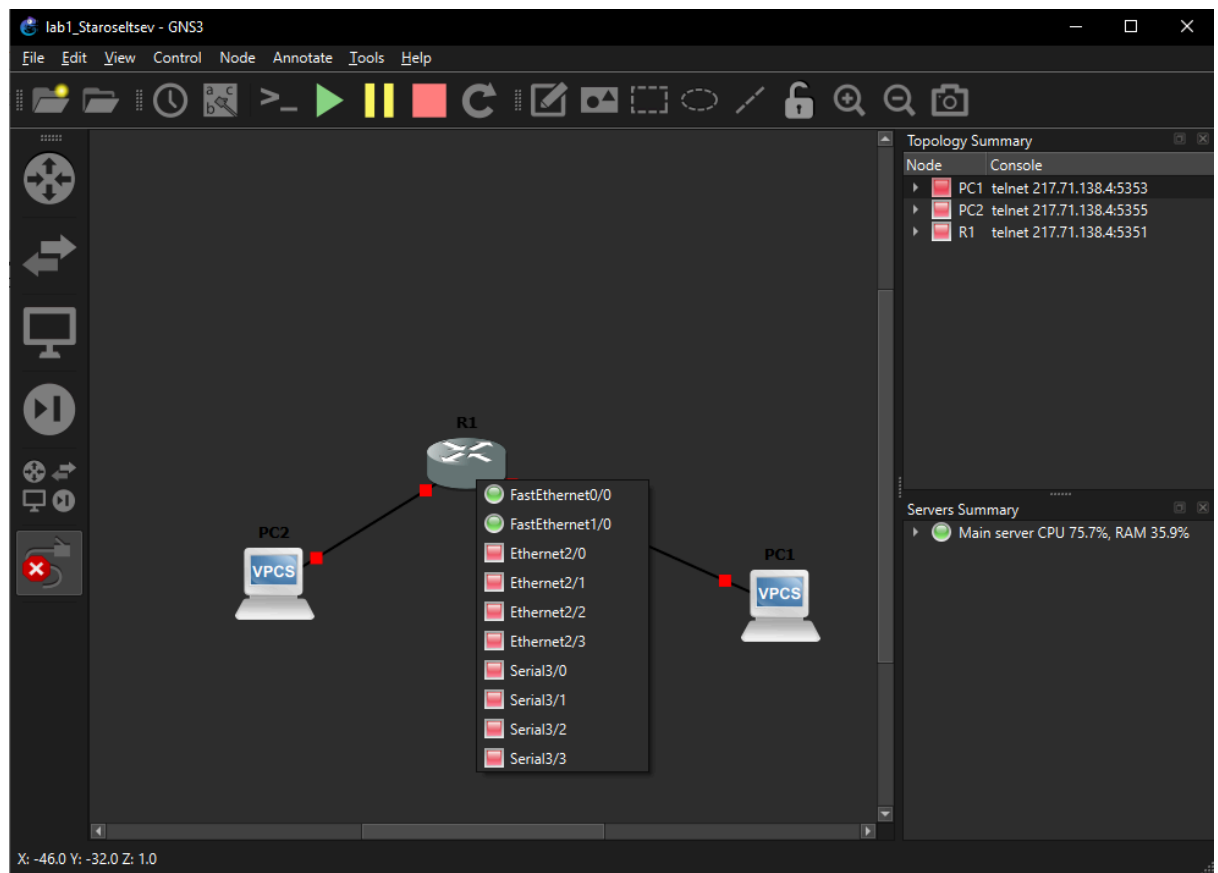
Ethernet II:
Destination MAC: 00:50:79:66:68:00
Source MAC: 00:50:79:66:68:01
Type: ARP

Address Resolution Protocol:
Hardware type: Ethernet
Protocol type: IPv4
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: 00:50:79:66:68:01
Sender IP address: 192.168.10.2
Target MAC address: 00:50:79:66:68:00
Target IP address: 192.168.10.1

При выполнении ping с PC1 на PC2 перед отправкой ICMP-пакетов был выполнен ARP-запрос. PC1 не знал MAC-адрес узла с IP 192.168.10.2, поэтому отправил широковещательный ARP Request с destination MAC ff:ff:ff:ff:ff:ff. Коммутатор переслал данный кадр на остальные порты, поэтому ARP Request был виден на всех линках топологии.

PC2 получил ARP Request и отправил ARP Reply, в котором сообщил свой MAC-адрес. ARP Reply передавался как unicast-кадр в сторону PC1. После получения ARP Reply PC1 смог сформировать Ethernet-кадр для ICMP Echo Request и выполнить ping.

5. Собрал топологию и подключил компьютеры по FastEthernet



R1 0/0 -> PC1 e0

R1 0/1 -> PC2 e0

PC1

ip 192.168.10.2/24 192.168.10.1

show ip

```
PC1> ip 192.168.10.2/24 192.168.10.1
Checking for duplicate address...
PC1 : 192.168.10.2 255.255.255.0 gateway 192.168.10.1

PC1> show ip

NAME       : PC1[1]
IP/MASK    : 192.168.10.2/24
GATEWAY    : 192.168.10.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 21742
RHOST:PORT : 127.0.0.1:21743
MTU        : 1500
```

PC2

ip 192.168.20.2/24 192.168.20.1

show ip

```

PC2> ip 192.168.20.2/24 192.168.20.1
Checking for duplicate address...
PC2 : 192.168.20.2 255.255.255.0 gateway 192.168.20.1

PC2> show ip

NAME       : PC2[1]
IP/MASK     : 192.168.20.2/24
GATEWAY     : 192.168.20.1
DNS         :
MAC         : 00:50:79:66:68:01
LPORT      : 21744
RHOST:PORT  : 127.0.0.1:21745
MTU         : 1500

```

Настройка роутера

Первый интерфейс

configure terminal

interface FastEthernet0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

exit

```

R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface FastEthernet0/0
R1(config-if)#ip address 192.168.10.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#exit
*Mar  1 00:07:35.643: %LINK-3-UPDOWN: Interface FastEthernet0/0, changed state to up
*Mar  1 00:07:36.643: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
R1(config-if)#exit

```

И для второго интерфейса

interface FastEthernet1/0

ip address 192.168.20.1 255.255.255.0

no shutdown

exit

end

```

R1(config)#interface FastEthernet1/0
R1(config-if)#ip address 192.168.20.1 255.255.255.0
R1(config-if)#no shutdown
R1(config-if)#
*Mar  1 00:10:17.007: %LINK-3-UPDOWN: Interface FastEthernet1/0, changed state to up
*Mar  1 00:10:18.007: %LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet1/0, changed state to up
R1(config-if)#exit

```

```

R1(config)#end
R1#
*Mar  1 00:11:10.295: %SYS-5-CONFIG_I: Configured from console by console

```

show ip interface brief


```

R1#show ip interface brief

```

Interface	IP-Address	OK?	Method	Status	Prot
FastEthernet0/0	192.168.10.1	YES	manual	up	up
FastEthernet1/0	192.168.20.1	YES	manual	up	up
Ethernet2/0	unassigned	YES	unset	administratively down	down
Ethernet2/1	unassigned	YES	unset	administratively down	down
Ethernet2/2	unassigned	YES	unset	administratively down	down
Ethernet2/3	unassigned	YES	unset	administratively down	down
Serial3/0	unassigned	YES	unset	administratively down	down
Serial3/1	unassigned	YES	unset	administratively down	down
Serial3/2	unassigned	YES	unset	administratively down	down
Serial3/3	unassigned	YES	unset	administratively down	down

6. Пинг с PC1 до PC2
ping 192.168.20.2

```

PC1> ping 192.168.20.2

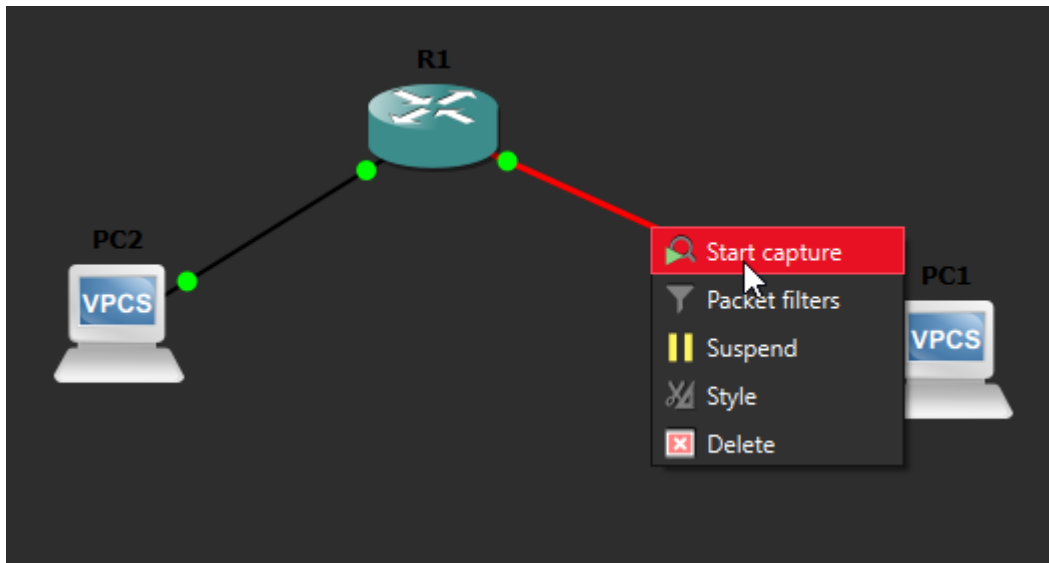
192.168.20.2 icmp_seq=1 timeout
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=17.643 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=11.383 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=21.856 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=18.018 ms

PC1> ping 192.168.20.2

84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=20.813 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=10.653 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=10.431 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=19.629 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=15.417 ms

```

7. Запускаем Wireshark на обоих линках



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	cc:01:2d:ea:00:00	cc:01:2d:ea:00:00	CDP/VTP/DTP/PagP/UD...	347	Device ID: R1 Port ID: FastEthernet0/0
2	6.205461	cc:01:2d:ea:00:00	cc:01:2d:ea:00:00	LOOP	60	Reply
3	16.202670	cc:01:2d:ea:00:00	cc:01:2d:ea:00:00	LOOP	60	Reply
4	26.198958	cc:01:2d:ea:00:00	cc:01:2d:ea:00:00	LOOP	60	Reply

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	cc:01:2d:ea:00:10	cc:01:2d:ea:00:10	LOOP	60	Reply
2	9.997011	cc:01:2d:ea:00:10	cc:01:2d:ea:00:10	LOOP	60	Reply
3	20.002825	cc:01:2d:ea:00:10	cc:01:2d:ea:00:10	LOOP	60	Reply

Очищаем кэш ARP

```
PC1> clear arp
PC2> clear arp
PC1>
PC2>
```

clear arp

И на роутере

```
Serial1/3/3
R1#clear arp-cache
R1#
```

clear arp-cache

Пингуем с PC1 PC2

ping 192.168.20.2

```
PC1> ping 192.168.20.2
84 bytes from 192.168.20.2 icmp_seq=1 ttl=63 time=10.801 ms
84 bytes from 192.168.20.2 icmp_seq=2 ttl=63 time=18.540 ms
84 bytes from 192.168.20.2 icmp_seq=3 ttl=63 time=19.717 ms
84 bytes from 192.168.20.2 icmp_seq=4 ttl=63 time=18.521 ms
84 bytes from 192.168.20.2 icmp_seq=5 ttl=63 time=17.222 ms
```

Захваченные пакеты:

ARP с линка PC1 R1

The image shows a Wireshark packet capture window titled "Захват с Standard input [PC1 Ethernet0 to R1 FastEthernet0/0]". The filter bar shows "arp". The packet list displays two packets:

No.	Time	Source	Destination	Protocol	Length	Info
3	15.614166	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.10.1? Tell 192.168.10.2
4	15.618194	cc:01:2d:ea:00:00	Private_66:68:00	ARP	60	192.168.10.1 is at cc:01:2d:ea:00:00

The packet details pane shows the selected packet (Frame 3) with the following structure:

- Frame 3: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0
- Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 - Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 - Source: Private_66:68:00 (00:50:79:66:68:00)
 - Type: ARP (0x0806)
 - [Stream index: 2]
 - Padding: 00000000000000000000000000000000
 - Frame check sequence: 0x00000000 [unverified]
 - [FCS Status: Unverified]
- Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: Private_66:68:00 (00:50:79:66:68:00)
 - Sender IP address: 192.168.10.2
 - Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 - Target IP address: 192.168.10.1

The packet bytes pane shows the raw data in hexadecimal and ASCII format.

Address Resolution Protocol: Protocol Пакеты: 30 · Отображено: 2 (6.7%) Профиль: Default

ICMP с линка PC1 R1

Захват с Standard input [PC1 Ethernet0 to R1 FastEthernet0/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

icmp

No.	Time	Source	Destination	Protocol	Length	Info
5	15.618404	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x3f4c, seq=1/256, ttl=64 (reply
6	15.648423	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x3f4c, seq=1/256, ttl=63 (reque
8	16.650709	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x404c, seq=2/512, ttl=64 (reply
9	16.667676	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x404c, seq=2/512, ttl=63 (reque
10	17.668025	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x414c, seq=3/768, ttl=64 (reply
11	17.685555	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x414c, seq=3/768, ttl=63 (reque
12	18.685750	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x424c, seq=4/1024, ttl=64 (repl
13	18.705188	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x424c, seq=4/1024, ttl=63 (requ
14	19.705730	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x434c, seq=5/1280, ttl=64 (repl
15	19.724327	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x434c, seq=5/1280, ttl=63 (requ

Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00)

- Destination: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00)
- Source: Private_66:68:00 (00:50:79:66:68:00)
- Type: IPv4 (0x0800)
- [Stream index: 3]

Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2

- 0100 = Version: 4
- 0101 = Header Length: 20 bytes (5)
- Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
- Total Length: 84
- Identification: 0x4c3f (19519)
- 000. = Flags: 0x0
- ...0 0000 0000 0000 = Fragment Offset: 0
- Time to Live: 64
- Protocol: ICMP (1)
- Header Checksum: 0x8f15 [validation disabled]
- [Header checksum status: Unverified]
- Source Address: 192.168.10.2
- Destination Address: 192.168.20.2
- [Stream index: 0]

0000 cc 01 2d ea 00 00
0010 00 54 4c 3f 00 00
0020 14 02 08 00 e0 be
0030 0e 0f 10 11 12 13
0040 1e 1f 20 21 22 23
0050 2e 2f 30 31 32 33
0060 3e 3f

Internet Control Message Protocol: Protocol

Пакеты: 39 · Отображено: 10 (25.6%)

Профиль: Default

ARP с линка PC2 R1

Захват с Standard input [PC2 Ethernet0 to R1 FastEthernet1/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

arp

No.	Time	Source	Destination	Protocol	Length	Info
4	19.390472	Private_66:68:01	Broadcast	ARP	64	Who has 192.168.20.1? Tell 192.168.20.2
5	19.400468	cc:01:2d:ea:00:10	Private_66:68:01	ARP	60	192.168.20.1 is at cc:01:2d:ea:00:10

Frame 4: Packet, 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface -, id 0

- Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
 - Destination: Broadcast (ff:ff:ff:ff:ff:ff)
 - Source: Private_66:68:01 (00:50:79:66:68:01)
 - Type: ARP (0x0806)
 - [Stream index: 2]
 - Padding: 00000000000000000000000000000000
 - Frame check sequence: 0x00000000 [unverified]
 - [FCS Status: Unverified]
- Address Resolution Protocol (request)
 - Hardware type: Ethernet (1)
 - Protocol type: IPv4 (0x0800)
 - Hardware size: 6
 - Protocol size: 4
 - Opcode: request (1)
 - Sender MAC address: Private_66:68:01 (00:50:79:66:68:01)
 - Sender IP address: 192.168.20.2
 - Target MAC address: Broadcast (ff:ff:ff:ff:ff:ff)
 - Target IP address: 192.168.20.1

0000 ff ff ff ff ff
0010 08 00 06 04 00
0020 ff ff ff ff ff
0030 00 00 00 00 00

Address Resolution Protocol: Protocol

Пакеты: 32 · Отображено: 2 (6.3%)

Профиль: Default

ICMP с линка PC2 R1

Захват с Standard input [PC2 Ethernet0 to R1 FastEthernet1/0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

icmp

No.	Time	Source	Destination	Protocol	Length	Info
3	19.390378	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x3f4c, seq=1/256
6	19.401266	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x3f4c, seq=1/256
8	20.419607	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x404c, seq=2/512
9	20.419714	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x404c, seq=2/512
10	21.437574	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x414c, seq=3/768
11	21.437636	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x414c, seq=3/768
12	22.457222	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x424c, seq=4/1024
13	22.457306	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x424c, seq=4/1024
14	23.476256	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x434c, seq=5/1280
15	23.476314	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x434c, seq=5/1280

Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)

Destination: Private_66:68:01 (00:50:79:66:68:01)

Source: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)

Type: IPv4 (0x0800)

[Stream index: 1]

Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2

0100 = Version: 4

.... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x4c3f (19519)

000. = Flags: 0x0

...0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 63

Protocol: ICMP (1)

Header Checksum: 0x9015 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.10.2

Destination Address: 192.168.20.2

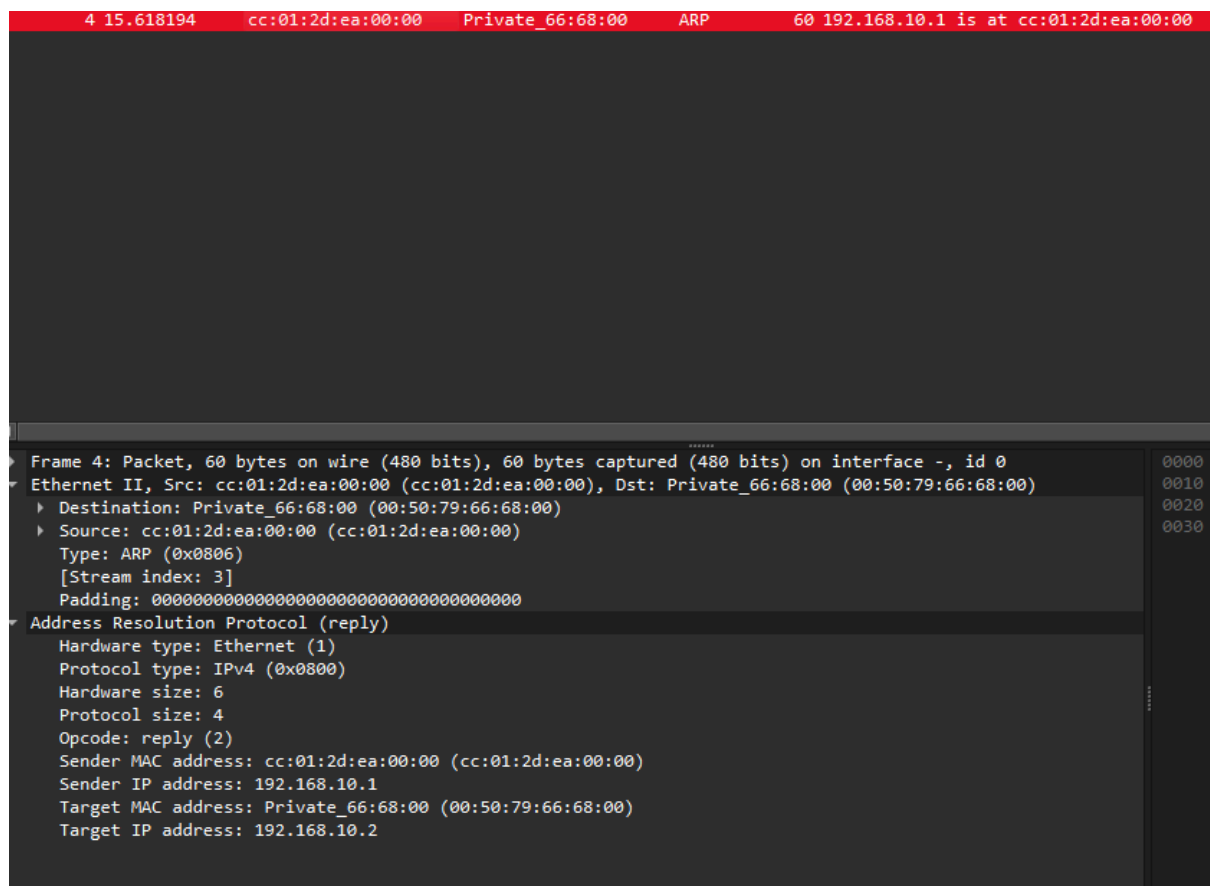
Internet Control Message Protocol: Protocol

Пакеты: 41 · Отображено: 10 (24.4%)

Профиль: Default

Анализ ARP

Первый линк:



Ethernet II:

Destination MAC: ff:ff:ff:ff:ff:ff

Source MAC: 00:50:79:66:68:00

Type: ARP (0x0806)

Поля ARP заголовка

Hardware type: Ethernet (1)

Protocol type: IPv4 (0x0800)

Hardware size: 6

Protocol size: 4

Opcode: request (1)

Sender MAC address: 00:50:79:66:68:00

Sender IP address: 192.168.10.2

Target MAC address: ff:ff:ff:ff:ff:ff

Target IP address: 192.168.10.1

В ответе R1 сообщает свой MAC

cc:01:2d:ea:00:00

Ethernet II:

Destination MAC: 00:50:79:66:68:00

Source MAC: cc:01:2d:ea:00:00

Type: ARP (0x0806)

Поля ARP заголовка

Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: reply (2)
Sender MAC address: cc:01:2d:ea:00:00
Sender IP address: 192.168.10.1
Target MAC address: 00:50:79:66:68:00
Target IP address: 192.168.10.2

Второй линк:

Запрос

The image shows a Wireshark packet capture interface. At the top, a packet list shows two packets: packet 4 (19.390472) is an ARP request from Private_66:68:01 to Broadcast, and packet 5 (19.400468) is an ARP reply from Private_66:68:01 to Private_66:68:01. The packet details pane for packet 4 is expanded, showing the Ethernet II header (Destination: Broadcast, Source: Private_66:68:01) and the ARP request header (Hardware type: Ethernet, Protocol type: IPv4, Opcode: request, Sender MAC address: Private_66:68:01, Sender IP address: 192.168.20.2, Target MAC address: Broadcast, Target IP address: 192.168.20.1). The packet bytes pane shows the raw data of the packet.

Компьютер PC2 с IP 192.168.20.2 ищет MAC-адрес своего шлюза 192.168.20.1, то есть интерфейса маршрутизатора R1. Так как MAC-адрес шлюза ещё неизвестен, PC2 отправляет ARP Request широковещательно. После этого маршрутизатор R1 отвечает ARP Reply и сообщает свой MAC-адрес cc:01:2d:ea:00:10.

Ethernet II:

Destination MAC: ff:ff:ff:ff:ff:ff
Source MAC: 00:50:79:66:68:01

Type: ARP (0x0806)

Поля ARP заголовка

Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4
Opcode: request (1)
Sender MAC address: 00:50:79:66:68:01
Sender IP address: 192.168.20.2
Target MAC address: ff:ff:ff:ff:ff:ff
Target IP address: 192.168.20.1

Ответ

5 19.400468 cc:01:2d:ea:00:10 Private_66:68:01 ARP 60 192.168.20.1 is at cc:01:2d:ea:00:10

Frame 5: Packet, 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface -, id 0

Ethernet II, Src: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)

- Destination: Private_66:68:01 (00:50:79:66:68:01)
- Source: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)
- Type: ARP (0x0806)
- [Stream index: 1]
- Padding: 00

Address Resolution Protocol (reply)

- Hardware type: Ethernet (1)
- Protocol type: IPv4 (0x0800)
- Hardware size: 6
- Protocol size: 4
- Opcode: reply (2)
- Sender MAC address: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)
- Sender IP address: 192.168.20.1
- Target MAC address: Private_66:68:01 (00:50:79:66:68:01)
- Target IP address: 192.168.20.2

0000 00 50 79 66
0010 08 00 06 04
0020 00 50 79 66
0030 00 00 00 00

Ethernet II:

Destination MAC: 00:50:79:66:68:01
Source MAC: cc:01:2d:ea:00:10
Type: ARP (0x0806)

Поля ARP заголовка

Hardware type: Ethernet (1)
Protocol type: IPv4 (0x0800)
Hardware size: 6
Protocol size: 4

Opcode: reply (2)
Sender MAC address: cc:01:2d:ea:00:10
Sender IP address: 192.168.20.1
Target MAC address: 00:50:79:66:68:01
Target IP address: 192.168.20.2

На втором линке PC2—R1 протокол ARP использовался для определения MAC-адреса шлюза по умолчанию. PC2 находится в сети 192.168.20.0/24 и для связи с другой сетью должен отправлять пакеты через маршрутизатор R1. Поэтому PC2 отправил широковещательный ARP-запрос для IP-адреса 192.168.20.1. Маршрутизатор R1 ответил ARP Reply и сообщил свой MAC-адрес cc:01:2d:ea:00:10. После этого PC2 получил возможность отправлять кадры на маршрутизатор для дальнейшей маршрутизации трафика.

Анализ ICMP:
Первый линк

The screenshot displays the Wireshark network protocol analyzer interface. The top pane shows a list of 15 ICMP Echo (ping) packets. The middle pane shows the details of the selected packet (Frame 5), which is an ICMP Echo request. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
5	15.618404	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x3f4c, seq=1/256, ttl=64 (reply in 6)
6	15.648423	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x3f4c, seq=1/256, ttl=63 (request in 5)
8	16.650709	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x404c, seq=2/512, ttl=64 (reply in 9)
9	16.667676	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x404c, seq=2/512, ttl=63 (request in 8)
10	17.668025	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x414c, seq=3/768, ttl=64 (reply in 11)
11	17.685555	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x414c, seq=3/768, ttl=63 (request in 10)
12	18.685750	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x424c, seq=4/1024, ttl=64 (reply in 13)
13	18.705188	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x424c, seq=4/1024, ttl=63 (request in 12)
14	19.705730	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request id=0x434c, seq=5/1280, ttl=64 (reply in 15)
15	19.724327	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply id=0x434c, seq=5/1280, ttl=63 (request in 14)

Frame 5: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00)

Destination: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00)

Source: Private_66:68:00 (00:50:79:66:68:00)

Type: IPv4 (0x0800)

[Stream index: 3]

Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2

0100 ... = Version: 4

... 0101 = Header Length: 20 bytes (5)

Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)

Total Length: 84

Identification: 0x4c3f (19519)

000. = Flags: 0x0

... 0 0000 0000 0000 = Fragment Offset: 0

Time to Live: 64

Protocol: ICMP (1)

Header Checksum: 0x8f15 [validation disabled]

[Header checksum status: Unverified]

Source Address: 192.168.10.2

Destination Address: 192.168.20.2

[Stream index: 0]

Internet Control Message Protocol: Protocol

Пакеты: 167 · Отображено: 10 (6.0%)

Профиль: Default

Echo Request: 192.168.10.2 → 192.168.20.2

Echo Reply: 192.168.20.2 → 192.168.10.2

PC1 отправляет ICMP Echo Request на PC2. Так как PC2 находится в другой сети, PC1 передаёт Ethernet-кадр не напрямую PC2, а на MAC-адрес маршрутизатора R1. При этом IP-адрес назначения остаётся 192.168.20.2

Запрос

```
Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
Ethernet II, Src: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)
  Destination: Private_66:68:01 (00:50:79:66:68:01)
  Source: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)
  Type: IPv4 (0x0800)
  [Stream index: 1]
Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 84
  Identification: 0x4c3f (19519)
  000. .... = Flags: 0x0
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 63
  Protocol: ICMP (1)
  Header Checksum: 0x9015 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.10.2
  Destination Address: 192.168.20.2
  [Stream index: 0]
Internet Control Message Protocol
  Type: Echo (ping) request (8)
  Code: 0
  Checksum: 0xe0be [correct]
  [Checksum Status: Good]
  Identifier (BE): 16204 (0x3f4c)
  Identifier (LE): 19519 (0x4c3f)
  Sequence Number (BE): 1 (0x0001)
  Sequence Number (LE): 256 (0x0100)
  [Response frame: 6]
  Data (56 bytes)
```

Ethernet II:

Destination MAC: cc:01:2d:ea:00:00

Source MAC: 00:50:79:66:68:00

Type: IPv4 (0x0800)

Internet Protocol Version 4:

Source IP address: 192.168.10.2

Destination IP address: 192.168.20.2

Header length: 20 bytes

Total length: 84

Time to live: 64

Protocol: ICMP (1)

Internet Control Message Protocol:

Type: 8 — Echo Request

Code: 0

Identifier: 0x3f4c

Sequence number: 1

PC1 отправляет ping-запрос к PC2. На IP-уровне получателем является 192.168.20.2, но на канальном уровне кадр отправляется на MAC-адрес интерфейса R1, потому что PC2 находится в другой сети.

Ответ

```
Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0
  Ethernet II, Src: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00), Dst: Private_66:68:00 (00:50:79:66:68:00)
    Destination: Private_66:68:00 (00:50:79:66:68:00)
    Source: cc:01:2d:ea:00:00 (cc:01:2d:ea:00:00)
    Type: IPv4 (0x0800)
    [Stream index: 3]
  Internet Protocol Version 4, Src: 192.168.20.2, Dst: 192.168.10.2
    0100 .... = Version: 4
    .... 0101 = Header Length: 20 bytes (5)
    Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 84
    Identification: 0x4c3f (19519)
    000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 63
    Protocol: ICMP (1)
    Header Checksum: 0x9015 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.20.2
    Destination Address: 192.168.10.2
    [Stream index: 0]
  Internet Control Message Protocol
    Type: Echo (ping) reply (0)
    Code: 0
    Checksum: 0xe8be [correct]
    [Checksum Status: Good]
    Identifier (BE): 16204 (0x3f4c)
    Identifier (LE): 19519 (0x4c3f)
    Sequence Number (BE): 1 (0x0001)
    Sequence Number (LE): 256 (0x0100)
    [Request frame: 5]
    [Response time: 30,019 ms]
  Data (56 bytes)
```

Ethernet II:

Destination MAC: 00:50:79:66:68:00

Source MAC: cc:01:2d:ea:00:00

Type: IPv4 (0x0800)

Internet Protocol Version 4:

Source IP address: 192.168.20.2

Destination IP address: 192.168.10.2

Header length: 20 bytes

Total length: 84

Time to live: 63

Protocol: ICMP (1)

Internet Control Message Protocol:

Type: 0 — Echo Reply

Code: 0

Identifier: 0x3f4c

Sequence number: 1

PC2 отвечает на ping-запрос. Ответ возвращается через маршрутизатор R1. На первом линке Ethernet-кадр идёт от MAC-адреса R1 к MAC-адресу PC1, но IP-источником остаётся 192.168.20.2, а IP-получателем — 192.168.10.2.

На линке PC1—R1 был зафиксирован ICMP-трафик, возникший при выполнении команды ping с PC1 на PC2. ICMP Echo Request имел IP-адрес источника 192.168.10.2 и IP-адрес назначения 192.168.20.2. Так как PC2 находится в другой сети, Ethernet-кадр был отправлен на MAC-адрес маршрутизатора R1 cc:01:2d:ea:00:00.

Второй линк:

Echo Request: 192.168.10.2 → 192.168.20.2

Echo Reply: 192.168.20.2 → 192.168.10.2

Запрос

icmp						
No.	Time	Source	Destination	Protocol	Length Info	
→	3 19.390378	192.168.10.2	192.168.20.2	ICMP	98 Echo (ping) request	id=0x3f4c, seq=1/256, ttl=63 (reply in 6)
←	6 19.401266	192.168.20.2	192.168.10.2	ICMP	98 Echo (ping) reply	id=0x3f4c, seq=1/256, ttl=64 (request in 3)
	8 20.419607	192.168.10.2	192.168.20.2	ICMP	98 Echo (ping) request	id=0x404c, seq=2/512, ttl=63 (reply in 9)
	9 20.419714	192.168.20.2	192.168.10.2	ICMP	98 Echo (ping) reply	id=0x404c, seq=2/512, ttl=64 (request in 8)
	10 21.437574	192.168.10.2	192.168.20.2	ICMP	98 Echo (ping) request	id=0x414c, seq=3/768, ttl=63 (reply in 11)
	11 21.437636	192.168.20.2	192.168.10.2	ICMP	98 Echo (ping) reply	id=0x414c, seq=3/768, ttl=64 (request in 10)
▶ Frame 3: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0						
▼ Ethernet II, Src: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10), Dst: Private_66:68:01 (00:50:79:66:68:01)						
▶ Destination: Private_66:68:01 (00:50:79:66:68:01)						
▶ Source: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)						
Type: IPv4 (0x0800)						
[Stream index: 1]						
▼ Internet Protocol Version 4, Src: 192.168.10.2, Dst: 192.168.20.2						
0100 = Version: 4						
.... 0101 = Header Length: 20 bytes (5)						
▶ Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)						
Total Length: 84						
Identification: 0x4c3f (19519)						
▶ 000. = Flags: 0x0						
...0 0000 0000 0000 = Fragment Offset: 0						
Time to Live: 63						
Protocol: ICMP (1)						
Header Checksum: 0x9015 [validation disabled]						
[Header checksum status: Unverified]						
Source Address: 192.168.10.2						
Destination Address: 192.168.20.2						
[Stream index: 0]						
▼ Internet Control Message Protocol						
Type: Echo (ping) request (8)						
Code: 0						
Checksum: 0xe0be [correct]						
[Checksum Status: Good]						
Identifier (BE): 16204 (0x3f4c)						
Identifier (LE): 19519 (0x4c3f)						
Sequence Number (BE): 1 (0x0001)						
Sequence Number (LE): 256 (0x0100)						
[Response frame: 6]						
▶ Data (56 bytes)						

Ethernet II:

Destination MAC: 00:50:79:66:68:01

Source MAC: cc:01:2d:ea:00:10

Type: IPv4 (0x0800)

Internet Protocol Version 4:

Source IP address: 192.168.10.2

Destination IP address: 192.168.20.2

Header length: 20 bytes

Total length: 84

Time to live: 63

Protocol: ICMP (1)

Internet Control Message Protocol:

Type: 8 — Echo Request

Code: 0

Identifier: 0x3f4c

Sequence number: 1

Маршрутизатор R1 пересылает ICMP Echo Request от PC1 к PC2.

IP-адрес источника остаётся 192.168.10.2, IP-адрес назначения остаётся 192.168.20.2.

Но на втором линке Ethernet-кадр уже идёт от MAC-адреса интерфейса R1 к MAC-адресу PC2.

Ответ:

icmp							
No.	Time	Source	Destination	Protocol	Length	Info	
→	3 19.390378	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request	id=0x3f4c, seq=1/256, ttl=63 (reply in 6)
←	6 19.401266	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply	id=0x3f4c, seq=1/256, ttl=64 (request in 3)
	8 20.419607	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request	id=0x404c, seq=2/512, ttl=63 (reply in 9)
	9 20.419714	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply	id=0x404c, seq=2/512, ttl=64 (request in 8)
	10 21.437574	192.168.10.2	192.168.20.2	ICMP	98	Echo (ping) request	id=0x414c, seq=3/768, ttl=63 (reply in 11)
	11 21.437636	192.168.20.2	192.168.10.2	ICMP	98	Echo (ping) reply	id=0x414c, seq=3/768, ttl=64 (request in 10)
Frame 6: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0							
Ethernet II, Src: Private_66:68:01 (00:50:79:66:68:01), Dst: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)							
Destination: cc:01:2d:ea:00:10 (cc:01:2d:ea:00:10)							
Source: Private_66:68:01 (00:50:79:66:68:01)							
Type: IPv4 (0x0800)							
[Stream index: 1]							
Internet Protocol Version 4, Src: 192.168.20.2, Dst: 192.168.10.2							
0100 = Version: 4							
.... 0101 = Header Length: 20 bytes (5)							
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)							
Total Length: 84							
Identification: 0x4c3f (19519)							
000. = Flags: 0x0							
...0 0000 0000 0000 = Fragment Offset: 0							
Time to Live: 64							
Protocol: ICMP (1)							
Header Checksum: 0x8f15 [validation disabled]							
[Header checksum status: Unverified]							
Source Address: 192.168.20.2							
Destination Address: 192.168.10.2							
[Stream index: 0]							
Internet Control Message Protocol							
Type: Echo (ping) reply (0)							
Code: 0							
Checksum: 0xe8be [correct]							
[Checksum Status: Good]							
Identifier (BE): 16204 (0x3f4c)							
Identifier (LE): 19519 (0x4c3f)							
Sequence Number (BE): 1 (0x0001)							
Sequence Number (LE): 256 (0x0100)							
[Request frame: 3]							
[Response time: 10,888 ms]							
Data (56 bytes)							

Ethernet II:

Destination MAC: cc:01:2d:ea:00:10

Source MAC: 00:50:79:66:68:01

Type: IPv4 (0x0800)

Internet Protocol Version 4:

Source IP address: 192.168.20.2

Destination IP address: 192.168.10.2

Header length: 20 bytes

Total length: 84

Time to live: 64

Protocol: ICMP (1)

Internet Control Message Protocol:

Type: 0 — Echo Reply

Code: 0

Identifier: 0x3f4c

Sequence number: 1

PC2 отправляет ICMP Echo Reply обратно к PC1.

Так как PC1 находится в другой сети, PC2 отправляет кадр на MAC-адрес своего шлюза — интерфейса R1.

IP-адрес источника в ответе — 192.168.20.2, IP-адрес назначения — 192.168.10.2.